

# CS224 Object Oriented Programming and Design Methodologies Quiz 2

Object Oriented Programming Quiz Question (C++)

## Quiz Guidelines

You will attempt this quiz on your lab machines. Your submission should be a single `.cpp` file named `firstname lastname quiz2.cpp`. The file should contain all of the code and implementation of the quiz problems.

Accessing any site other than HULMS and the provided documentation is prohibited. You are not allowed to communicate with your peers or use any external resources. You cannot refer to your notes, books or lecture slides during the quiz. Use of generative AI is strictly prohibited. Not abiding by any of the rules will result in an immediate zero in the quiz grade and a possible academic conduct case.

## Scenario

With the recent increase in Devil sightings in Tokyo, Makima, the head of the Public Safety Devil Hunter Division, has decided that her loyal Devil Hunters require some “special training.” She tasks you with creating a turn-based battle simulation program in C++ using classes and objects, a request you can’t refuse.

### Part A: Class Move

Define a class `Move` with the following **private** attributes:

- **name**: a character array of size 50, storing the move’s name
- **type**: a character array of size 10, storing the move’s type (“Fire” or “Water”)
- **baseDamage**: an integer, storing the amount of damage the move deals
- **effect**: a character array of size 100, storing a description of the move’s special effect

### Methods:

- `getBaseDamage()`: returns `baseDamage`
- `getType()`: returns `type`
- `DisplayName()`: prints `<move name>`
- `DisplayEffect()`: prints `Special Effect: <effect>`

**Part B: Class Creature**

Define a class **Creature** with the following **private** attributes:

- **name**: a character array of size 50, storing the Creature's name
- **health**: an integer, storing the Creature's current health
- **type**: a character array of size 10, storing the Creature's type ("Fire" or "Water")
- **primaryAttack**: an object of type Move
- **secondaryAttack**: an object of type Move
- **boostCounter**: an integer

**Methods:**

1. **GetName()**: returns the Creature's name.
2. **GetHealth()**: returns the Creature's health.
3. **GetType()**: returns the Creature's type.
4. **SetBoost()**: increments **boostCounter**; resets it to 0 if greater than 3.
5. **GetBoost()**: returns the value of **boostCounter**.
6. **TakeDamage(int damage)**:
  - Reduces the Creature's **health** by **damage**.
7. **Attack(Creature &opponent)**:
  - (a) Prints the attacker's name and lists their attacks (Name, Attack Type, Base Damage).
  - (b) Prompts the user to choose an attack (1 or 2).
  - (c) Prints the following: <attackers name> used <selected move>
  - (d) Calls the move's **DisplayEffect()**.
  - (e) Using the moves type and the opponents type calculates type-based damage:
    - Fire (Move) vs Water (Opponent) → half damage.
    - Water (Move) vs Fire (Opponent) → double damage.
    - Same type → normal damage.
  - (f) Applies damage to the opponent using the **TakeDamage()** method.
  - (g) Prints opponent's name, the damage the opponent took and the opponents remaining health.
8. **Attack(Creature &opponent, int boostDamage)** (Overloaded):
  - Same as above, but adds **boostDamage** to the final damage.
  - Prints Boost Damage Applied: +<boostDamage>.

**Part C: main() Function**

The **main()** function has been provided to you. You do not need to write it.

## How to compile

You should be able to compile your code using the following terminal command:

```
g++ mainBattle.cpp Move.cpp Creature.cpp -o Battle
```

Followed by:

```
./Battle
```

## Sample Test Cases

### Test Case 1: Normal Attacks Only

```
>>> Denji's Turn!
>>> Choose attack:
1. Chainsaw (Fire, 50 dmg)
2. Beam (Water, 40 dmg)
>>> Enter choice: 2
Denji used Beam!
Special Effect: It shoots a stream of water!
Reze took 80 damage!
Reze's Health: 40

>>> Reze's Turn!
>>> Choose attack:
1. BOOM!! (Fire, 60 dmg)
2. Spark (Water, 45 dmg)
>>> Enter choice: 1
Reze used BOOM!!
Special Effect: It creates a colorful explosion!
Denji took 30 damage!
Denji's Health: 70

>>> Denji's Turn!
>>> Choose attack:
1. Chainsaw (Fire, 50 dmg)
2. Beam (Water, 40 dmg)
>>> Enter choice: 2
Denji used Beam!
Special Effect: It shoots a stream of water!
Reze took 80 damage!
```

Reze's Health: -40

=====

Winner: Denji (Remaining Health: 70)

=====

### Test Case 2: Boost Counter Activated

>>> Denji's Turn!

>>> Choose attack:

1. Chainsaw (Fire, 50 dmg)

2. Beam (Water, 40 dmg)

>>> Enter choice: 1

Denji used Chainsaw!

Special Effect: It engulfs the arena in flames!

Reze took 25 damage!

Reze's Health: 95

>>> Reze's Turn!

>>> Choose attack:

1. BOOM!! (Fire, 60 dmg)

2. Spark (Water, 45 dmg)

>>> Enter choice: 2

Reze used Spark!

Special Effect: Electric sparks fly across the battlefield!

Denji took 45 damage!

Denji's Health: 55

>>> Denji's Turn! (Boost counter reached 3)

>>> Choose attack:

1. Chainsaw (Fire, 50 dmg)

2. Beam (Water, 40 dmg)

>>> Enter choice: 2

Denji used Beam!

Special Effect: It shoots a stream of water!

Boost Damage Applied: +5

Reze took 85 damage!

Reze's Health: 10

```
>>> Reze's Turn!
>>> Choose attack:
1. BOOM!! (Fire, 60 dmg)
2. Spark (Water, 45 dmg)
>>> Enter choice: 1
Reze used BOOM!!
Special Effect: It creates a colorful explosion!
Denji took 30 damage!
Denji's Health: 25
```

```
>>> Denji's Turn!
>>> Choose attack:
1. Chainsaw (Fire, 50 dmg)
2. Beam (Water, 40 dmg)
>>> Enter choice: 2
Denji used Beam!
Special Effect: It shoots a stream of water!
Reze took 80 damage!
Reze's Health: -70
```

```
=====
Winner: Denji (Remaining Health: 25)
=====
```